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AI-Assisted Content and Language Integrated Learning (CLIL)

Enhancing English Language Proficiency among Tamil-Medium Social Sciences Undergraduates

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Introduction

- Content and Language Integrated Learning (CLIL) is widely recognised as an effective pedagogical approach for integrating disciplinary content with second language development.
- CLIL focuses on the successful acquisition of a new language through the processes of understanding and creating meaning.
- In CLIL, content and target-language acquisition are treated as being of equal importance, whereas in English Medium Instruction (EMI), content takes precedence over the medium through which it is delivered.
- The growing integration of artificial intelligence in higher education has opened new possibilities for enhancing both language learning and subject-content understanding through AI-assisted CLIL.
- This study investigates the effectiveness of AI-assisted CLIL in improving the English language proficiency of Tamil-medium, second-year Social Sciences undergraduates at the University of Jaffna, Sri Lanka.

Literature Context

- Unlike traditional approaches, the CLIL approach focuses on the creative and functional use of language for cognitive development and communication.
- Thompson and McKinley note that CLIL supports learners in achieving their situational language needs.
- In the Sri Lankan context, CLIL aligns with national educational goals of improving English proficiency for academic advancement and global opportunities.
- Research suggests that CLIL enhances learner engagement by providing comprehensible input, encouraging interaction, and facilitating meaningful output, fostering cognitive and linguistic development.
- CLIL also promotes learner motivation through authentic and relevant content, contributing to positive language-identity formation.
- AI integration offers additional support by providing adaptive input, scaffolding comprehension, and enhancing reading engagement.

Objective

To examine the effectiveness of AI-assisted CLIL in improving the L2 proficiency of second-year Tamil-medium Social Sciences undergraduates at the University of Jaffna, Sri Lanka.

Research Hypothesis

Students' performance in L2 is significantly higher after the AI-assisted CLIL intervention.

Methodology

Design

One-group quasi-experimental, mixed-methods design

Participants

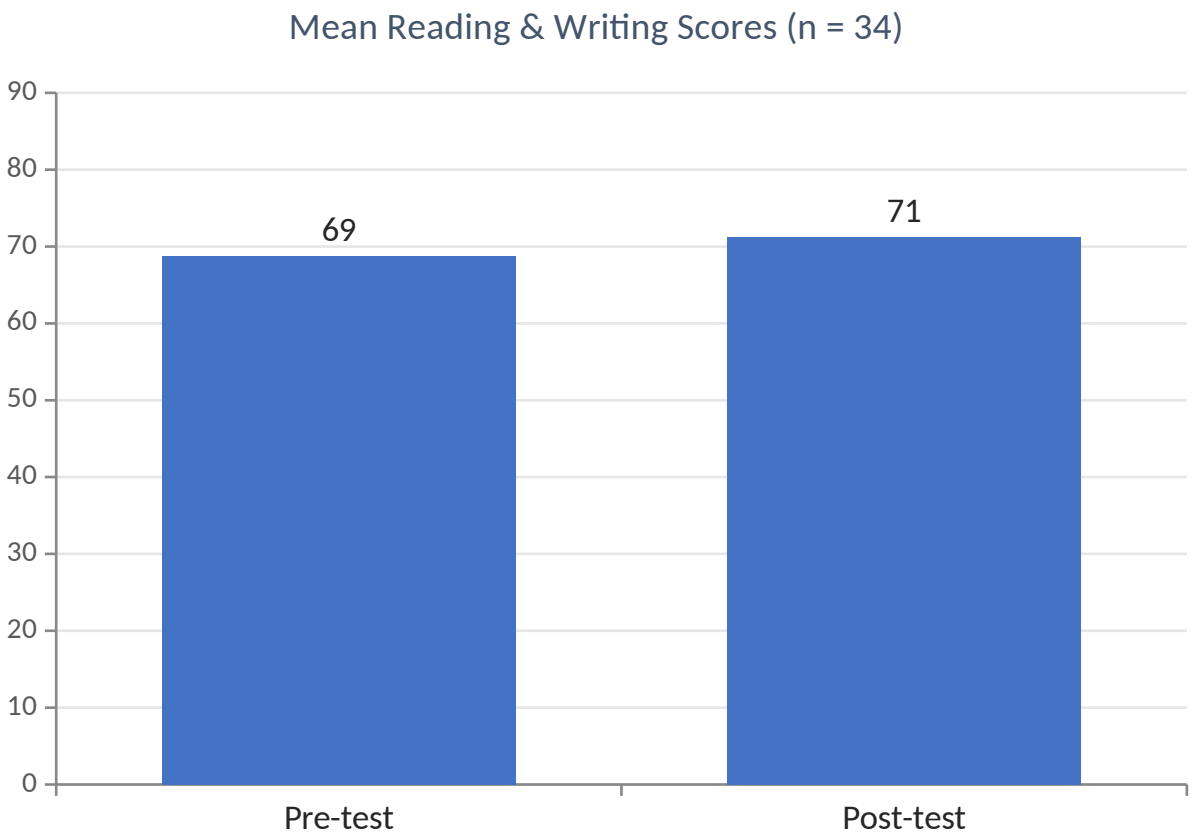
34 second-year Tamil-medium Social Sciences undergraduates, one academic semester

Data Collection

Pre-test, post-test, and semi-structured interviews (ethical approval and informed consent obtained)

- Students completed researcher-developed pre-test and post-test assessments measuring English reading and writing skills.
- Semi-structured interviews were conducted to explore learners' perceptions of, and experiences with, the intervention.
- Participants engaged with Social Sciences-related English texts and tasks while using AI tools such as ChatGPT and Gemini to support content and vocabulary development and text comprehension.
- Weekly presentations were also held to develop speaking fluency, although this fell outside the primary research focus.

Findings: Pre-test vs. Post-test Scores



Pre-test (M, SD)	**68.76, 10.74**
Post-test (M, SD)	**71.18, 11.57**
Paired t-test	**t(33) = 2.68, p = .011**
Effect size	**Cohen's d = 0.46 (moderate)**
Students who improved	**23 of 34 (67.6%)**

Post-test scores were significantly higher than pre-test scores, indicating a statistically significant and educationally meaningful improvement in students' English reading and writing performance following the AI-assisted CLIL intervention.

Findings: Learner Perceptions

- Participants perceived AI-assisted CLIL as conducive to independent learning.
- All participants reported increased confidence in using, and an improved ability to access and understand, materials in English.
- The majority of participants also reported a growing reliance on AI tools for self-directed learning.
- Participants' speaking fluency in relation to subject-related topics also appeared to improve.
- Taken together, these findings suggest that AI plays a valuable role in supporting self-directed learning within a CLIL framework.

Conclusion

- AI-assisted CLIL is an effective pedagogical approach for promoting English language development while supporting disciplinary knowledge and learner autonomy.
- As the majority of students reported a degree of over-reliance on AI for self-directed learning, there is a clear need for facilitators to teach Critical AI Literacy.
- Critical AI Literacy training should equip students to recognise bias, identify misinformation, and use AI tools ethically.
- Future research could investigate the long-term impact of AI-assisted CLIL across different disciplines, using larger samples and longitudinal research designs.

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THANK YOU

Questions & Discussion